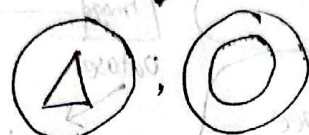


Line Drawing Algo + Circle Drawing Algo + $\Delta, \square$

$$y = mx + c$$

x					
y					

$$a^x + b^y = n^z$$



Board $\rightarrow$ Uniform
Pixel $\rightarrow$ Uniform not

• Line + Circle $\rightarrow$ 2D, 3D

• color model $\rightarrow$ $\begin{matrix} R & G & B \\ \downarrow & \downarrow & \downarrow \\ 255 & 255 & 255 \end{matrix}$

• shading

• Transformation $\rightarrow$ $\begin{matrix} \square & \square & \square \\ \square & \square & \square \end{matrix}$

Lecture-01 (Introduction to computer Graphics)

Graphics Systems and Model

Applications of Computer Graphics

• Computer Aided Decisions

$\rightarrow$ CAD: Shading, lighting, Rendering

$\rightarrow$ Blender: How to make donuts

• Presentation Graphics

$\rightarrow$ Powerpoint, Excel

• computer Art

$\rightarrow$ Photoshop, Blender

• Entertainment $\rightarrow$ Pixar, Avatars

- Education & Training

→ DIU Logo,

- Visualization

→ Bar chart, Box plot

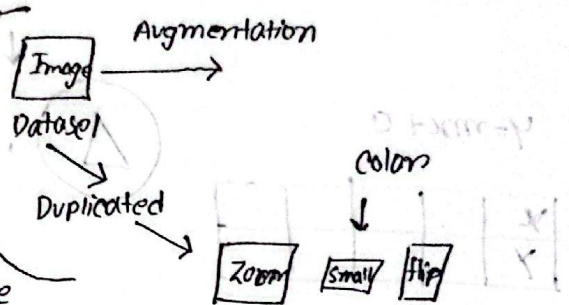
- Image Processing

→ contrast ↑

- Graphical User Interface

→ Simple features

→ Good Graphical Experience



(Week-02, L-03) Lecture-03 (Line Drawing Algorithm)

371 Algorithm

(i) DDA Line Drawing Algorithm

① Starting & Ending Coordinates

② Slope: $\frac{\Delta y}{\Delta x}$

→ Given 1. Start Point, End Point (2,5) (5,8)

2. Slope Calculation, $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

3. After finding the value of slope we determine ~~whether~~ ^{how} we increase x and y

Case 1: $0 < m \leq 1$ (Ex: $\frac{1}{2}$) অংশে $x_1 = x_0 + 1, y_1 = y_0 + m$

Case 2: $-1 < m \leq 0$ (Ex: 0) অংশে $x_1 = x_0 - 1, y_1 = y_0 - m$

Case 3: $m > 1$ (Ex: 2) অংশে $x_1 = x_0 + 1/m, y_1 = y_0 + 1$

Case 4: $m \leq -1$ (Ex: -2) অংশে $x_1 = x_0 - 1/m, y_1 = y_0 - 1$

Problem 01:Ans: $(5, 6) \equiv (x_0, y_0)$

slope, $M = \frac{12-6}{8-5} = \frac{6}{3} = 2$

 $(8, 12) \equiv (x_n, y_n)$ $\therefore$ Num. of required ^{iterations} equation, $\text{Max}(\Delta x, \Delta y) = \text{Max}(3, 6) = 6$ $\therefore$ Slope $M = 2 > 1 \therefore x_1 = x_0 + \frac{1}{M}, y_1 = y_0 + 1$

	x_p	y_p	x_{p+1}	y_{p+1}	Round Off (x_{p+1}, y_{p+1})
1.	5	6	$5 + \frac{1}{2} = 5.5$	7	(6 , 7)
2.			6	8	(6 , 8)
3.			6.5	9	(7 , 9)
4.			7	10	(7 , 10)
5.			7.5	11	(8 , 11)
6.			8	12	(8 , 12)

মসল $m=1$ আসবে, এককমাত্র এই case এই straight line পাবে।Problem 02: $(5, 6), (13, 10) \therefore M = \frac{10-6}{13-5} = \frac{4}{8} = 0.5 \therefore \text{iterations} = 8$ Slope, $M = 0.5 \leq 1 \therefore x_1 = x_0 + 1, y_1 = y_0 + M$

	x_p	y_p	x_{p+1}	y_{p+1}	Round Off
	5	6	6	$6 + 0.5 = 6.5$	(6 , 7)
			7	$6.5 + 0.5 = 7$	(7 , 7)
			8	7.5	(8 , 8)
			9	8	(9 , 8)
			10	8.5	(10 , 9)
			11	9	(11 , 9)
			12	9.5	(12 , 10)
			13	10	(13 , 10)

$6.5 = 7$

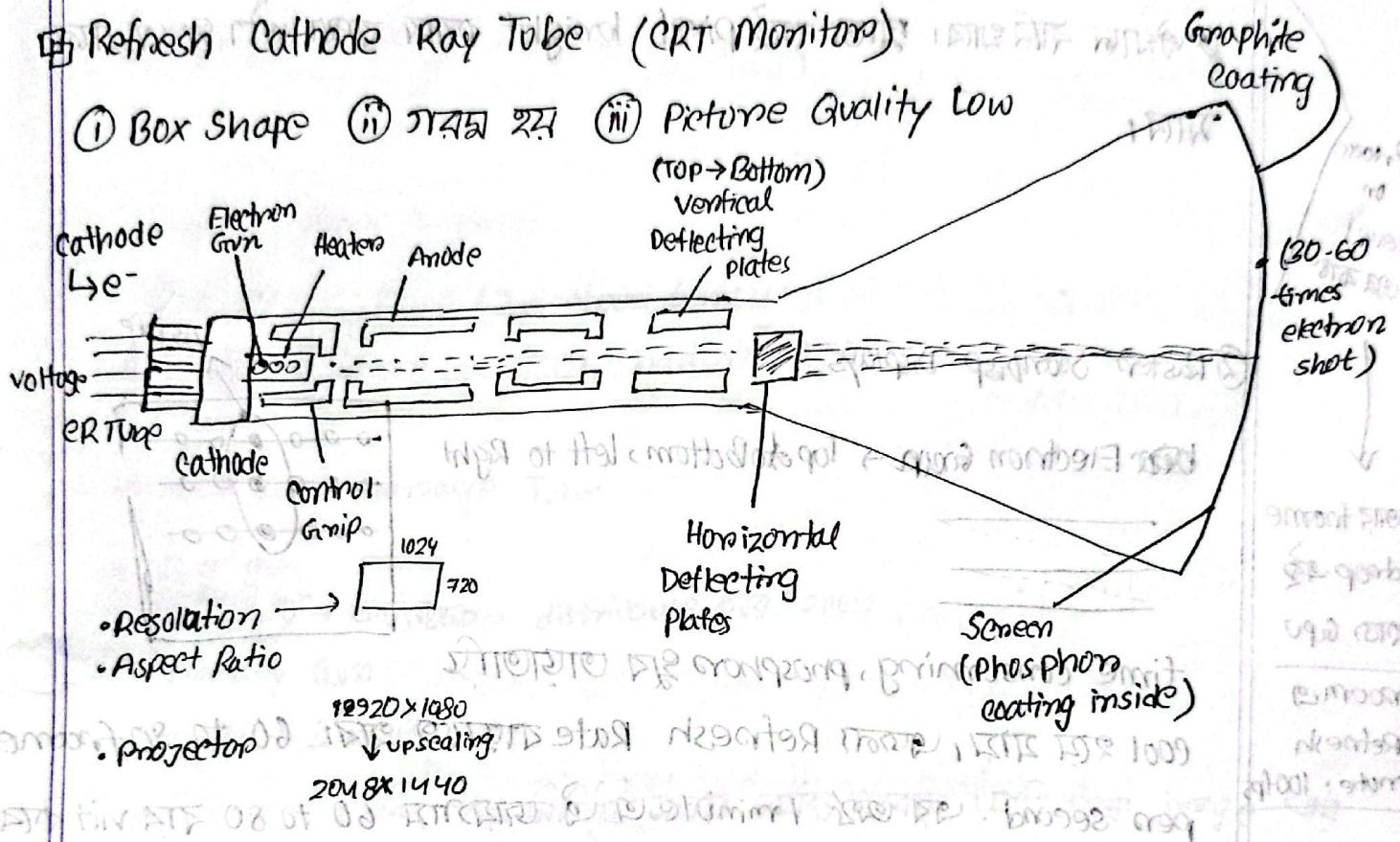
$6.4 = 6$

Quiz ৩ আসবে
case 2, 4Desmos
for graph

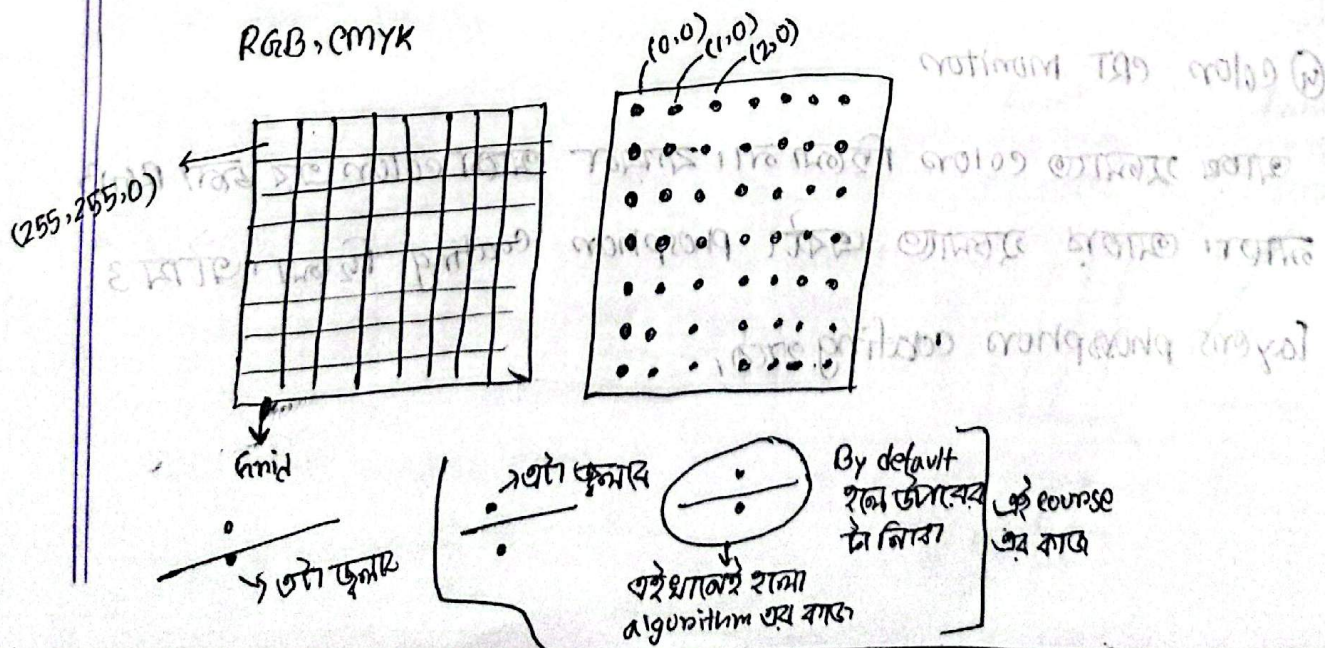
Lecture-02 (Video Display Devices)

Refresh Cathode Ray Tube (CRT Monitor)

- ① Box shape ② গরম হয় ③ Picture Quality Low



③ Random Scan Image is a matrix • Pixel যুক্ত। এর

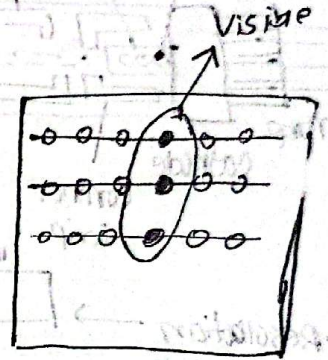
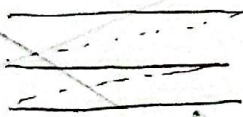


③ Random Scan Display

(মেথানেই drawing শুরু হবে ~~and beam~~ এখানেই থাকবে, দ্বারা e- মনো
এখানে বারিধাব। দ্বারা মেই pixel) bright করার ব্যাধ। এটা ~~এ~~ এখানে
মার।

③ Raster Scan Displays

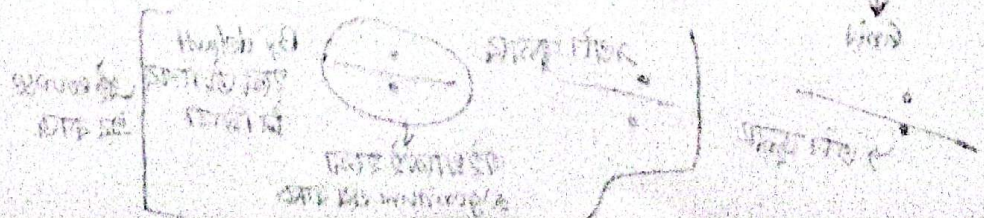
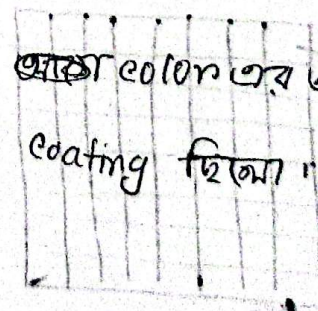
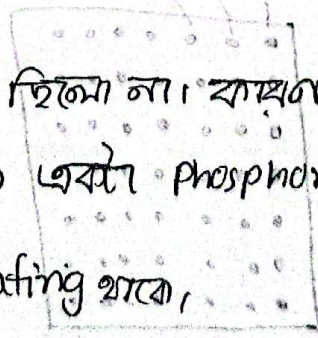
~~Electron~~ Electron Beam → Top to Bottom, left to Right



time consuming, phosphor খুব তাড়াতাড়ি
cool হয়ে যায়, এজন্য Refresh Rate বাড়তে হবে। 60 to 80 frame
per second. এর অর্থ 1 minute এ ~~এ~~ জায়গায় 60 to 80 বার দিত করে
Unit: Hertz (Hz)

④ Color CRT Monitor

আজ যুগ্মতে color ছিলা না। কারণ ~~এ~~ color এর জন্য Red
না। আজ যুগ্মতে একটা phosphor coating ছিলা। এমই 3
layers phosphor coating থাকে,



Screen
on
level
এর কাছ

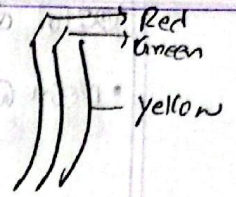
আর frame
drop এই
কাজে GPU
room এ
Refresh
rate: 100fps

আর RR;
58 fps

4.1 Beam - Penetration Method

৭টা color দেখাতে পারবে।

phosphor coating 2 layer



4.2 Shadow - Mask Method

→ Red, Green, Blue (3 electron guns)

→ phosphor screen এর আঁচর shadow থাকবে।

5 Direct View Storage Tube

→ 2টি e-gun

(Flood Gun) → আনব্রুজ - distribute করে open

Primary Gun

→ only Black & white

→ একটি image তৈরি সব নির্দিষ্ট সময় পর্যন্ত থাকবে। যদি স্ক্রিন image এর ক্ষয় হলে অস্পষ্ট কাটতে চাই তাহলে দুই image করেও হবে।

6 Flat Panel Display

→ Glass concept

6.1 Plasma Panel

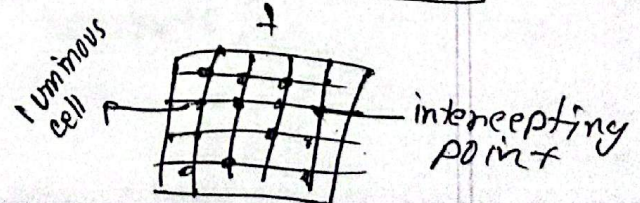
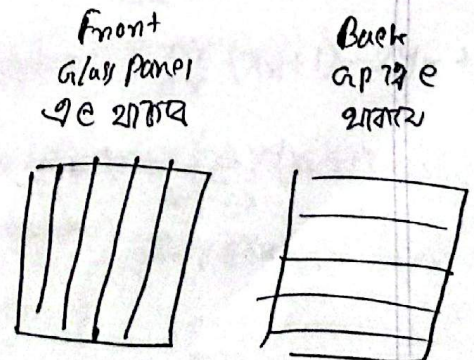
- layers of glasses as like sandwich

- Dielectric layers এ থাকবে -

- দ্বারা থাকবে plasma cell

- neon gas ও stored থাকে দ্বারা

↓
inert gas: Xenon (তাহলে e-কে hit করে।)



display & light show করে।

- Neon Gas হলো atoms দিয়ে গঠিত। এ কারণে ওয়াহাটোইটি করে।

Advantages:

Disadvantages:

(6.2) Thin-Film Electroluminescent

neon gas X

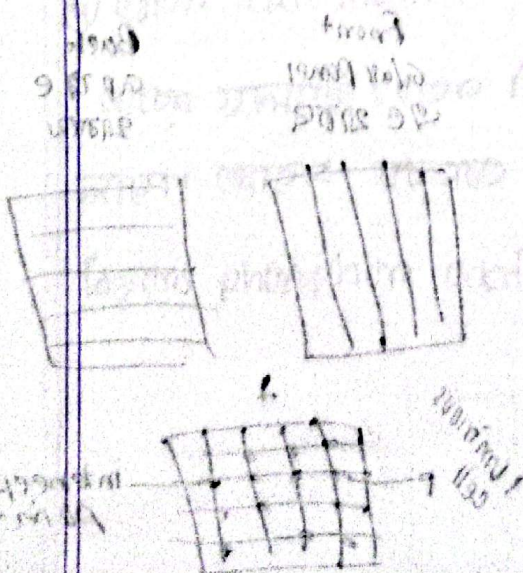
L-phospho Layer ✓

(6.3) LED

(6.4) LAI

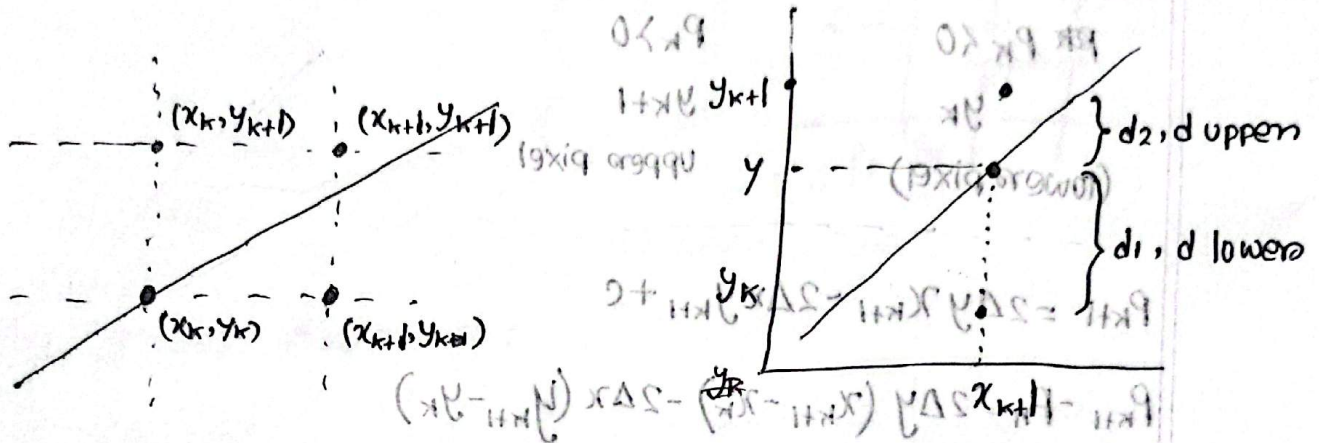
> কোন situation এ কোনটা use করবে।
Advantage, Disadvantage কোনটার কি

important for theory part in mid



Slide even .app

Bresenham's Line Algorithm



$$y = mx + b$$

slope, $m \leq 1$ and $m > 1$

$$\begin{aligned} y_{k+1} &\rightarrow y = d \text{ upper} \\ y &\rightarrow y_k = d \text{ lower} \end{aligned}$$

Given,

straight line, $y = mx + c$

$$\text{slope, } m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx} = 2b - 1 + 2m = 2y_k + 2mx_k = 2m(x_k + 1) - 2y_k + 2b - 1$$

for slope $m \leq 1$

x will be always incremented

$$x_{k+1}$$

Substitute, $m = \frac{\Delta y}{\Delta x}$

$$d_{\text{lower}} - d_{\text{upper}} = 2 \frac{\Delta y}{\Delta x} (x_{k+1}) - 2y_k + 2b - 1$$

$$y = m(x_k + 1) + b$$

$$d_{\text{upper}} = (y_{k+1}) - y$$

$$= y_{k+1} - m(x_k + 1) - b$$

$$d_{\text{lower}} = y - y_k$$

$$= m(x_k + 1) + b - y_k$$

Decision Parameter:

$$P_k = \Delta x (\text{dlower} - \text{dupper})$$

$$P_k < 0$$

$$y_k$$

(lower pixel)

$$P_k > 0$$

$$y_{k+1}$$

upper pixel

$$P_{k+1} = 2\Delta y x_{k+1} - 2\Delta x y_{k+1} + c$$

$$P_{k+1} - P_k = 2\Delta y (x_{k+1} - x_k) - 2\Delta x (y_{k+1} - y_k)$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x (y_{k+1} - y_k)$$

~~$P_0 = 2\Delta y \Delta x$~~ when $P_k < 0$ then dlower - dupper < 0

d lower $<$ d upper ; we choose the lower pixel

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x (y_k - y_k)$$

$$\therefore P_{k+1} = P_k + 2\Delta y$$

when $P_k > 0$ then dlower - dupper > 0

$\therefore$ d lower - dupper > 0 ; we will choose the upper pixel

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x (y_{k+1} - y_k)$$

$$P_{k+1} = 2\Delta y \cdot x_k - 2\Delta x y_k + c = 2\Delta y \cdot x_k - 2\Delta x y_k + \Delta x (2b - 1) + 2\Delta y$$

$$\therefore P_0 = 2\Delta y \cdot x_0 - 2\Delta x y_0 + c = 2\Delta y \cdot x_0 - 2\Delta x y_0 + \Delta x (2b - 1) + 2\Delta y$$

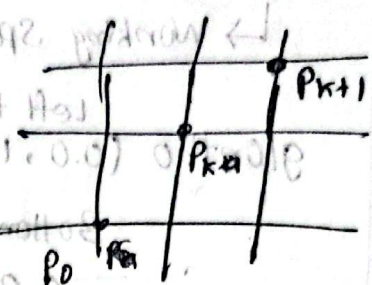
$$\therefore P_0 = 2\Delta y - \Delta x$$

formulas

$$P_0 = 2\Delta y - \Delta x$$

$$P_{k+1} = P_k + 2\Delta y$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$



Initial Decision Parameters:

$$P_0 = 2\Delta y - \Delta x$$

$$P_0 < 0$$

$$x_{k+1} = x_k + 1$$

$$P_{k+1} = P_k + 2\Delta y$$

$$y_{k+1} = y_k$$

$$P_0 > 0$$

$$x_{k+1} = x_k$$

$$y_{k+1} = y_k + 1$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$

14-06-2026

Bresenham's Line Drawing Algorithm:

Fall 2025

Derivation (must learn)

Start Point (x_0, y_0) → End Point (x_n, y_n)

① Find dy and dx (Δy and Δx)

$$dy = |y_n - y_0| \quad dx = |x_n - x_0|$$

② find slope.

$$m = \frac{dy}{dx}$$

③ find the change of x and y

④ If slope ≤ 1 $\therefore P_0 = 2dy - dx$

If $P_k \geq 0$: x_{k+1} (x change) $= x_k + 1$

y_{k+1} (y change) $= y_k$

$$P_{k+1} = P_k + 2dy - 2dx$$

If $P_k < 0$:

$x_{k+1} = x_k$ (x change)

$y_{k+1} = y_k + 1$ (y change)

$$P_{k+1} = P_k + 2dy$$

If slope > 1

$$P_0 = 2dx - dy$$

if $P_k > 0$

$x_{k+1} = x_k + 1$ (x change)

$y_{k+1} = y_k$ (y change)

$$P_{k+1} = P_k + 2dx - 2dy$$

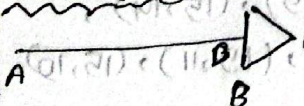
if $P_k < 0$

$x_{k+1} = x_k$

$y_{k+1} = y_k + 1$ (y change)

$$P_{k+1} = P_k + 2dx$$

Draw (ques):



- A (10, 10)
- B (15, 10)
- C (15, 15)
- D (15, 10)
- E (20, 10)

Ans: for AB, ①

A (10, 10) → B (15, 10)

$$\therefore dx = |15 - 10| = 5$$

$$\therefore dy = |10 - 10| = 0$$

$$\textcircled{2} \text{ Slope, } m = \frac{dy}{dx} = \frac{0}{5} = 0$$

$\therefore x \text{ change} = +1$

$y \text{ change} = 0$

$$\textcircled{4} \therefore P_0 = 2dy - dx = 2 \cdot 0 - 5 = -5$$

P_k	P_{k+1}	x	y
		10	10
-5	-5	$10+1=11$	10
-5	-5	$11+1=12$	10
-5	-5	13	10
-5	-5	14	10
-5	-5	15	10

Output Screen

∴ Drawing Pixels are:

$(10,10), (11,10), (12,10), (13,10), (14,10), (15,10)$

for ED :

$e(15,15) \rightarrow D(15,5)$

$$dx = |15-15| = 0$$

$$dy = |15-5| = 10$$

$$\text{slope } m = \frac{dy}{dx} = \frac{10}{0} = \infty$$

∴ $x \text{ change} = 0$

∴ $y \text{ change} = -1$

∴ Since $\text{slope} > 1$

$$P_0 = 2dx - dy = 2 \times 0 - 10 = -10$$

∴ Drawing Pixels are:

$(10,10), (11,10), (12,10), (13,10), (14,10), (15,10), (15,11), (15,12), (15,13), (15,14), (15,15)$

P_k	P_{k+1}	x	y
		15	15
-10	-10	15	$15-1=14$
-10	-10	15	13
-10	-10	15	12
-10	-10	15	11
-10	-10	15	10

for E,
 $E(15,15) \rightarrow E(20,10)$

$$\therefore dx = |15-20| = 5$$

$$\therefore dy = |15-10| = 5$$

$$\text{slope, } m = \frac{dy}{dx} = \frac{5}{5} = 1$$

$$\therefore x \text{ change} = +1$$

$$y \text{ change} = -1$$

$$P_0 = 2 \times 5 - 5 = 5$$

	P_k	P_{k+1}	x	y
			15	15
0	5	5	16	14
1	5	5	17	13
2	5	5	18	12
3	5	5	19	11
4	5		20	10

$\therefore$ Drawing Pixels are:

$(15,15), (16,14), (17,13), (18,12), (19,11), (20,10)$

Ques: A(9,3) B(6,8)

Apply Bresenham's Line Drawing Algorithm to draw AB line segment.

A(9,3) $\rightarrow$ B(6,8)

$$dx = |9-6| = 3$$

$$dy = |3-8| = 5$$

$$\therefore m = \frac{dy}{dx} = \frac{5}{3} = 1.667$$

$$\therefore x \text{ change} = -1$$

$$\therefore y \text{ change} = +1$$

$$\therefore P_0 = 2 \times 3 - 5 = 1$$

	P_k	P_{k+1}	x	y
			9	3
0	1	-3	8	4
1	-3	3	8	5
2	3	-1	7	6
3	-1	5	7	7
4	5		6	8

$\therefore$ Drawing Pixels are: $(9,3), (8,4), (8,5), (7,6), (7,7), (6,8)$

In code:

```
int x0 = 0
int y0 = 0
int xn = 10
int yn = 10
```

```
dy = abs(yn - y0)
dx = abs(xn - x0)
slope = dy / dx
```

```
if x0 > xn
    xchange = -1
else if x0 < xn
    xchange = 1
else
    xchange = 0
```

```
if y0 > yn
    ychange = -1
else
    ychange = 1
```

```
if y0 > yn
    ychange = -1
else if y0 < yn
    ychange = 1
else
    ychange = 0
```

```
if slope < 1
    while (xn != xk)
```

```
    p0 = 2 * dy - dx
    xk = x0, yk = y0, pk = p0
```

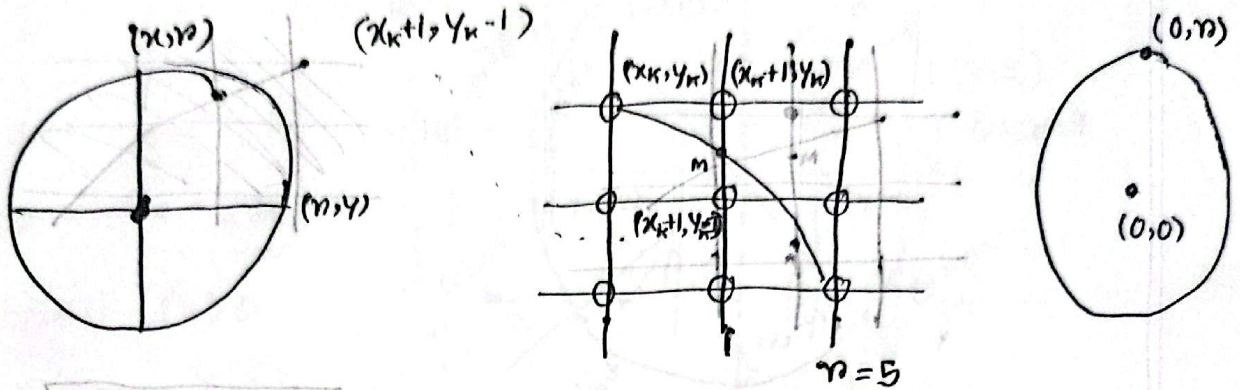
```
    if pk >= 0
        xk1 = xk + xchange
        yk1 = yk + ychange
```

```
    draw(xk1, yk1)
```

x	y
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10

...

Mid Point Circle Drawing Algorithm



Now, $x^2 + y^2 = r^2$ $f(x, y) = x^2 + y^2 - r^2$

$f(x, y) = 0$ (point is on the circle boundary)

> 0 (point is outside of circle boundary)

< 0 (point is inside of circle boundary)

$$\therefore \text{Midpoint } M = \left\{ \frac{x_k + x_{k+1}}{2}, \frac{y_k + y_{k+1}}{2} \right\} = \left\{ x_k + \frac{1}{2}, y_k + \frac{1}{2} \right\}$$

$$\text{Therefore, } f\left(x_k + \frac{1}{2}, y_k + \frac{1}{2}\right) = \left(x_k + \frac{1}{2}\right)^2 + \left(y_k + \frac{1}{2}\right)^2 - r^2$$

$$\therefore \text{Initial Decision Parameter } P_k = f\left(x_k + \frac{1}{2}, y_k + \frac{1}{2}\right) = \left(x_k + \frac{1}{2}\right)^2 + \left(y_k + \frac{1}{2}\right)^2 - r^2$$

$$\therefore P_{k+1} = \left(x_{k+1} + \frac{1}{2}\right)^2 + \left(y_{k+1} + \frac{1}{2}\right)^2 - r^2$$

যাহাৰে পিক্সেল ডিচিজন এ x এর value বাঢ়ে (যিহেতু $x_{k+1} = x_k + 1$)

$$\text{Now, } P_{k+1} = \left(x_k + 1 + \frac{1}{2}\right)^2 + \left(y_k + \frac{1}{2}\right)^2 - r^2 = \left(x_k + \frac{3}{2}\right)^2 + \left(y_k + \frac{1}{2}\right)^2 - r^2$$

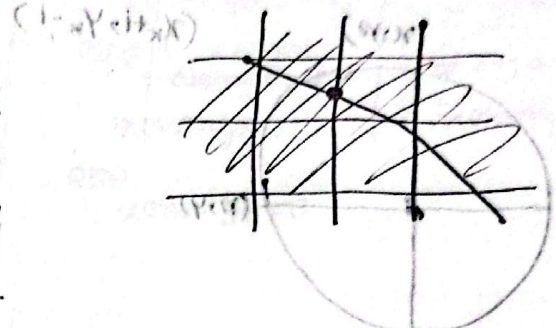
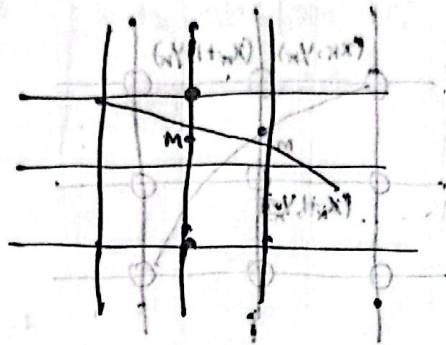
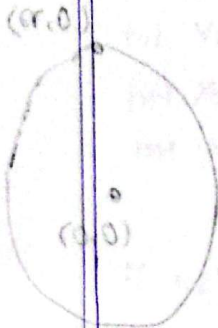
Now,

$$P_{k+1} - P_k = \left(x_k + \frac{3}{2}\right)^2 - \left(x_k + \frac{1}{2}\right)^2 + \left(y_k + \frac{1}{2}\right)^2 - \left(y_k + \frac{1}{2}\right)^2$$

$$= x_k^2 + 4x_k + \frac{9}{4} - x_k^2 - 2x_k - \frac{1}{4} + y_k^2 + y_k + \frac{1}{4} - y_k^2 - y_k + \frac{1}{4}$$

$$= 2x_k + 3 + (y_{k+1})^2 - y_{k+1} - (y_k)^2 + y_k$$

$$P_{k+1} = P_k + 2x_k + 3 + (y_{k+1})^2 - y_{k+1} - (y_k)^2 + y_k$$



if $P_k < 0$ $\therefore y_{k+1} = y_k$

if $P_k > 0$ $\therefore y_{k+1} = y_k - 1$

$$P_{k+1} = P_k + 2x_k + 3$$

(point is outside of circle boundary)

(point is inside of circle boundary)

$$\left\{ \frac{1}{2} - x_k + y_k \right\} = \left\{ \frac{1}{2} - x_k + y_k \right\}$$

$$P_{k+1} = P_k + 2(x_k + y_k) + 5$$

$$P_0 = (0+1)^2 + (0-\frac{1}{2})^2 - \frac{5}{4} = -\frac{1}{4}$$

$$P_{k+1} = (x_{k+1}+1)^2 + (y_{k+1}-\frac{1}{2})^2 - \frac{5}{4}$$

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$$P_{k+1} = (x_k+1)^2 + (y_k-\frac{1}{2})^2 - \frac{5}{4}$$

for calculation:

Draw a circle at center (6,7) with a diameter of 10

Step 1: starting point (0,0)

$P_0 = \frac{5}{4} - \frac{1}{4}$ (radius $\rightarrow$ int)

$P_0 = \frac{5}{4} - \frac{1}{4}$ (radius $\rightarrow$ float)

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Shafin-184

main

Step 2:

if $P_k < 0$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 2x_k + 3$$

else if $P_k > 0$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k - 1$$

$$P_{k+1} = P_k + 2(2x_k - y_k) + 5$$

stop: $x \geq y$

$$(F, 1) \leftarrow (0, 2)$$

$$(S, 2) \leftarrow (2, 0)$$

$$(F, 11) \leftarrow (0, 2) \quad (S, 11) \leftarrow (2, 0)$$

We will always calculate

Octant 2 (x, y)

$$(E, 11) \leftarrow (1, 2) \quad (S, 11) \leftarrow (2, 1)$$

$$\text{Octant 2} \rightarrow \text{Octant 1} \begin{matrix} (1, 5) \leftarrow (5, 1) \\ (11, 5) \leftarrow (5, 11) \end{matrix}$$

$$(E, 8) \leftarrow (11, 5) \quad (E, 9) \leftarrow (5, 11)$$

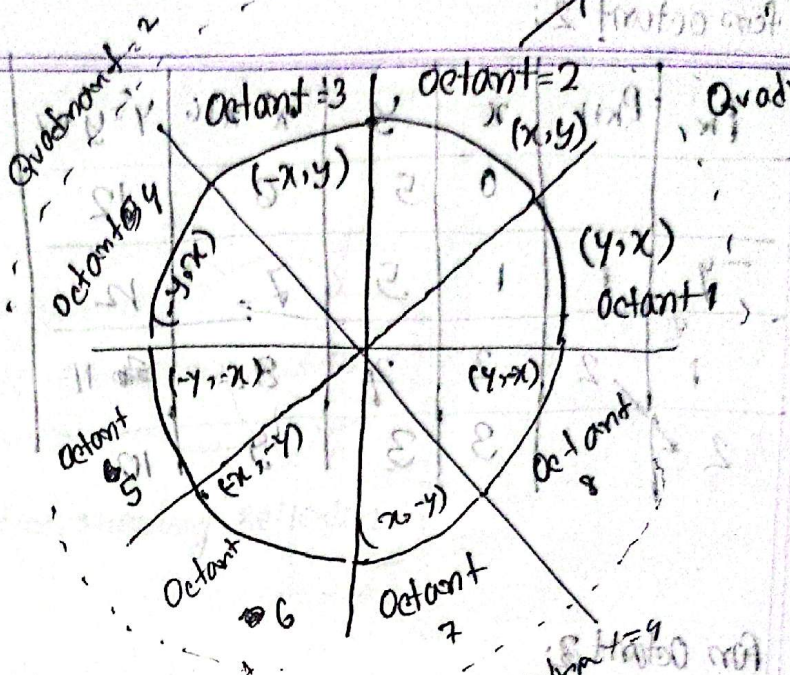
$$(E, 9) \leftarrow (5, 11) \quad (11, 8) \leftarrow (11, 5)$$

(x, y)

$$10, 15 \rightarrow (15, 10)$$

$$11, 19 \rightarrow (19, 11)$$

$$12, 13 \rightarrow (13, 12)$$



Ques: Draw a circle at center $(6, 7)$ with a diameter of 10

Ans: Given, center $(6, 7)$, Radius = $\frac{10}{2} = 5$. starting point $(0, 5)$

starting point $(0, 0)$

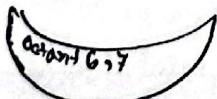
$$P_0 = 1 - r^2 = 1 - 5^2 = -24$$

for octant 2:

P_k	P_{k+1}	x	y	$x' = x + 6$	$y' = y + 7$
		0	5	6	12
$-y$		1	5	7	12
1	2	2	4	8	11
2		3	3	9	10

for octant 3:

x, y	y, x	$x, -y$	$-y, x$
Octant 2	Octant 1	Octant 7	Octant 4
$(0, 5) \rightarrow (6, 12)$	$(5, 0) \rightarrow (11, 7)$	$(0, -5) \rightarrow (6, 2)$	$(-5, 0) \rightarrow (11, 7)$
$(1, 5) \rightarrow (7, 12)$	$(5, 1) \rightarrow (11, 8)$	$(1, -5) \rightarrow (7, 2)$	$(-5, 1) \rightarrow (11, 8)$
$(2, 4) \rightarrow (8, 11)$	$(4, 2) \rightarrow (10, 9)$	$(2, -4) \rightarrow (8, 3)$	$(-4, 2) \rightarrow (2, 1)$
$(3, 3) \rightarrow (9, 10)$	$(3, 3) \rightarrow (9, 10)$	$(3, -3) \rightarrow (9, 4)$	$(-3, 3) \rightarrow (3, 10)$



Ques: Draw a circle of center (0,0) with a radius of 10.

Ans: Given, center (0,0). Radius = 10.

Point (0,0) is the center.

$$x^2 + y^2 = 10^2$$

Question Quiz-01: An Important Lecture on Bresenham's Line

Drawing Algorithm

Lecture-06

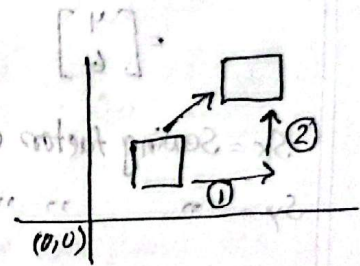
2D Transformation

Translation, Scaling, Rotation, Shearing, Reflexion

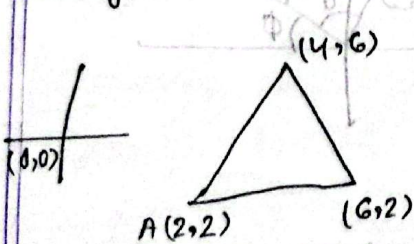
↓
অবস্থান slide দ্বারা নাহে

↓
2D Transformation (Translation)

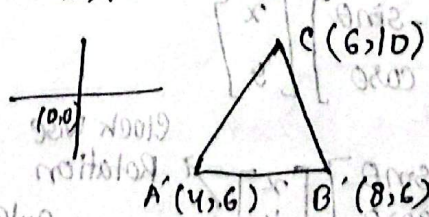
1st Obj ko निर्दिष्ट द्वारा নিয়ে যাওয়া হল Translation
যদি x-axis move এবং y-axis move



Ques: A triangle of points A(2,2), B(6,2), C(4,6) is moved 2 coordinates along x-axis and 4 coordinates along y-axis



So, $A(2,2) \rightarrow A'(x',y')$



Translation: for point A:

formula, $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$

Point B' = $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$

Point C' = $\begin{bmatrix} 6 \\ 10 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$

Scaling (Size ছোট বড় - Zoom in & Zoom Out):

Scaling factor $> 1 \Rightarrow$ Big size

Scaling factor $< 1 \Rightarrow$ Small size

$A(2,2)$, $S_x = 2$, $S_y = 3$

formula, $A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

S_x = Scaling factor of x axis

S_y = " " " y axis

Rotation:

~~$A(x,y)$~~ $A(x,y) \rightarrow \theta$

formula: $A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

formula: $A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Anti Clock Wise

(ঘড়ী না বন্ধা থাকে)

(কিছু ACW)

Clock Wise
Rotation

calculator Degree ত থাকবে

Example: $A(0,0)$ $B(4,4)$ $\theta = 30^\circ$

$$B' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos 30 & -\sin 30 \\ \sin 30 & \cos 30 \end{bmatrix} \times \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 1.46 \\ 5.46 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

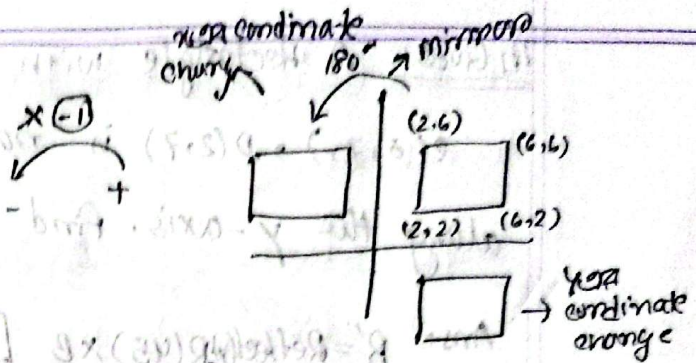
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Reflection:

① Reflection of X-Axis:

$A(x, y)$

$$A'(x', y') = \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$



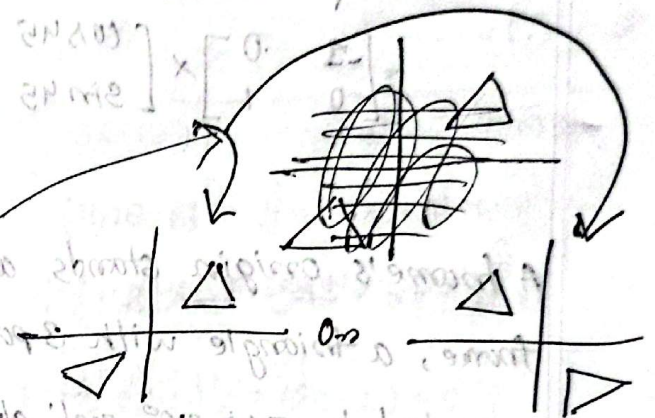
② Reflection of Y-Axis:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

③ Reflection of Origin:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

→ Reflection about an axis perpendicular to xy plane and passing through origin
→ It is also called half revolution about the origin.



Shearing → (Distorting objects along an axis) → Baseline change

Example:

$A \rightarrow (1, 1)$, $shx = 2$

$$A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$A \rightarrow (1, 1)$, $shy = 2$

$$A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

① Shearing on X-axis:

$A(x, y)$

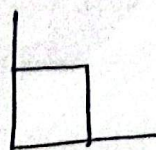
$$A' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & shx \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

② Shearing on Y-axis:

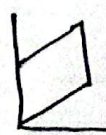
$B(x, y)$

$$B' \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ shy & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

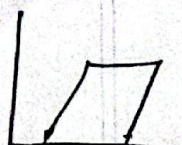
Graph-plot → Depends on the questions



normal



Y-axis shear



After X-axis shear

Ques: A Rectangle with four points A(2,2), B(8,2), C(8,7), D(2,7) is rotated by 45° and then reflected along the y-axis. Find the new points of the triangle.

Ans: $B' = \text{Reflect}(R(45)) \times B$ [Order 45 and reflect.]

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{bmatrix} \times \begin{bmatrix} 8 \\ 2 \end{bmatrix} = \begin{bmatrix} -4.25 \\ 7.07 \end{bmatrix}$$

A frame's origin stands at (2,3) with respect to the given frame, a triangle with 3 points A(2,2), B(6,2), C(4,6) is rotated by 56° anti clock wise direction and scaled is scaled by scaling parameter 2 on x-axis and scaling parameter 4 on y-axis. Find the new coordinates with respect to the equation

Ans: $C' = \text{Translate}(2,3)$

~~Translate~~ ~~(2,3)~~ $\times$ Scaling(2,4) $\times$ Rotation(56°) $\times$ Translate(-2,-3) $\times$ C

$$C', \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos 56 & -\sin 56 & 0 \\ \sin 56 & \cos 56 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 4 \\ 6 \\ 1 \end{bmatrix}$$

Confusion

0 or 1
maybe

Confusion

maybe 01

